Molecular Featurization II: Fingerprints

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University of South Carolina

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From Lecture 24

Molecular Featurization I: SMILES and RDKit

- *Read simple SMILES strings and inspect molecules with RDKit*
- *Represent SMILES tokens using one-hot vectors*
- *Construct and concatenate atom-count and bond-count vectors.*

In this lecture, you will learn

- *Morgan fingerprints and local atomic environments*
- *Generating fingerprints with RDKit*
- *Comparing molecules using fingerprint similarity*

Why Atom and Bond Counts Are Insufficient

Different molecules can share both atom counts and bond-type counts.

Molecule	SMILES	Atom counts [C, O]	Bond counts [C–C, C–O]
1-Propanol	`CCCO`	`[3, 1]`	`[2, 1]`
2-Propanol	`CC(O)C`	`[3, 1]`	`[2, 1]`

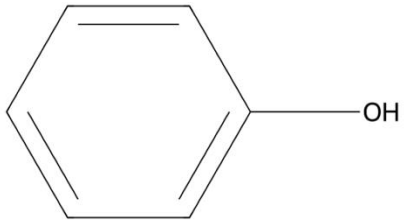
$$\mathbf{x} = [3, 1, 2, 1]$$

Bond-type counts retain some connectivity information, but they do not specify how all the bonds fit together.

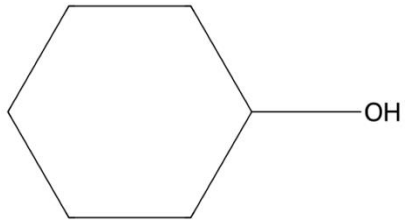
What feature could distinguish an oxygen attached to an end carbon from an oxygen attached to the middle carbon?

Which Candidate Resembles Phenol Most?

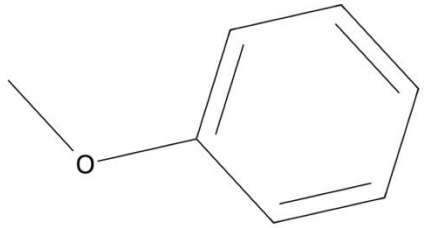
Reference: phenol



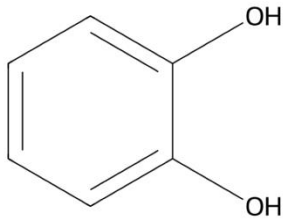
A: Cyclohexanol



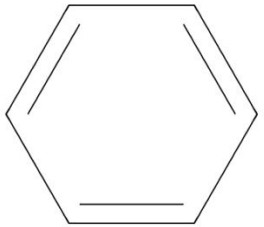
B: Anisole



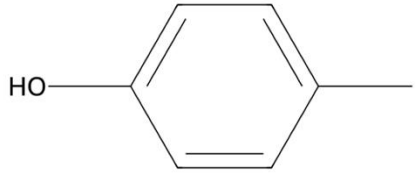
C: Catechol



D: Benzene



E: p-Cresol

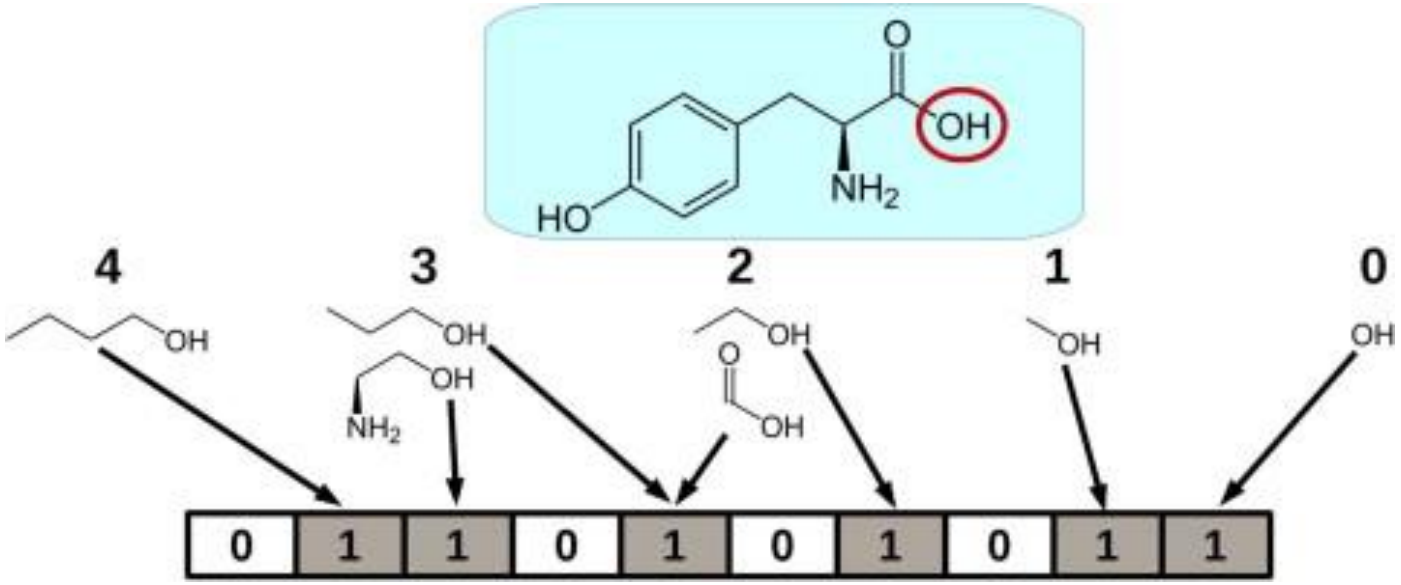


Molecular Fingerprint

A fingerprint encodes structural features in a numerical vector.

Binary fingerprints record feature presence.

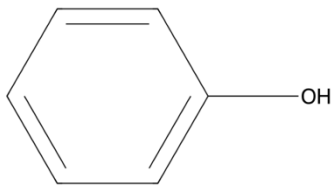
Count fingerprints record feature occurrences.



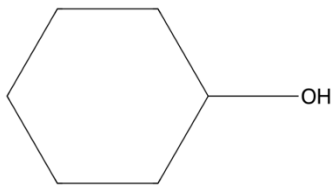
Cereto-Massagué, Adrià, et al. "Molecular fingerprint similarity search in virtual screening." Methods 71 (2015): 58-63.

Different Fingerprints Capture Different Features

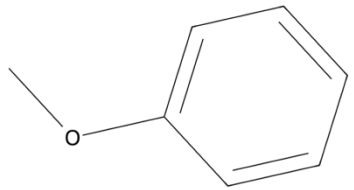
Reference: phenol



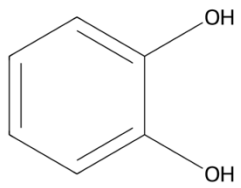
A: Cyclohexanol



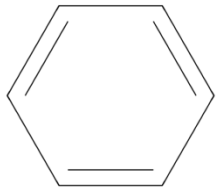
B: Anisole



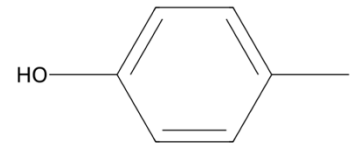
C: Catechol

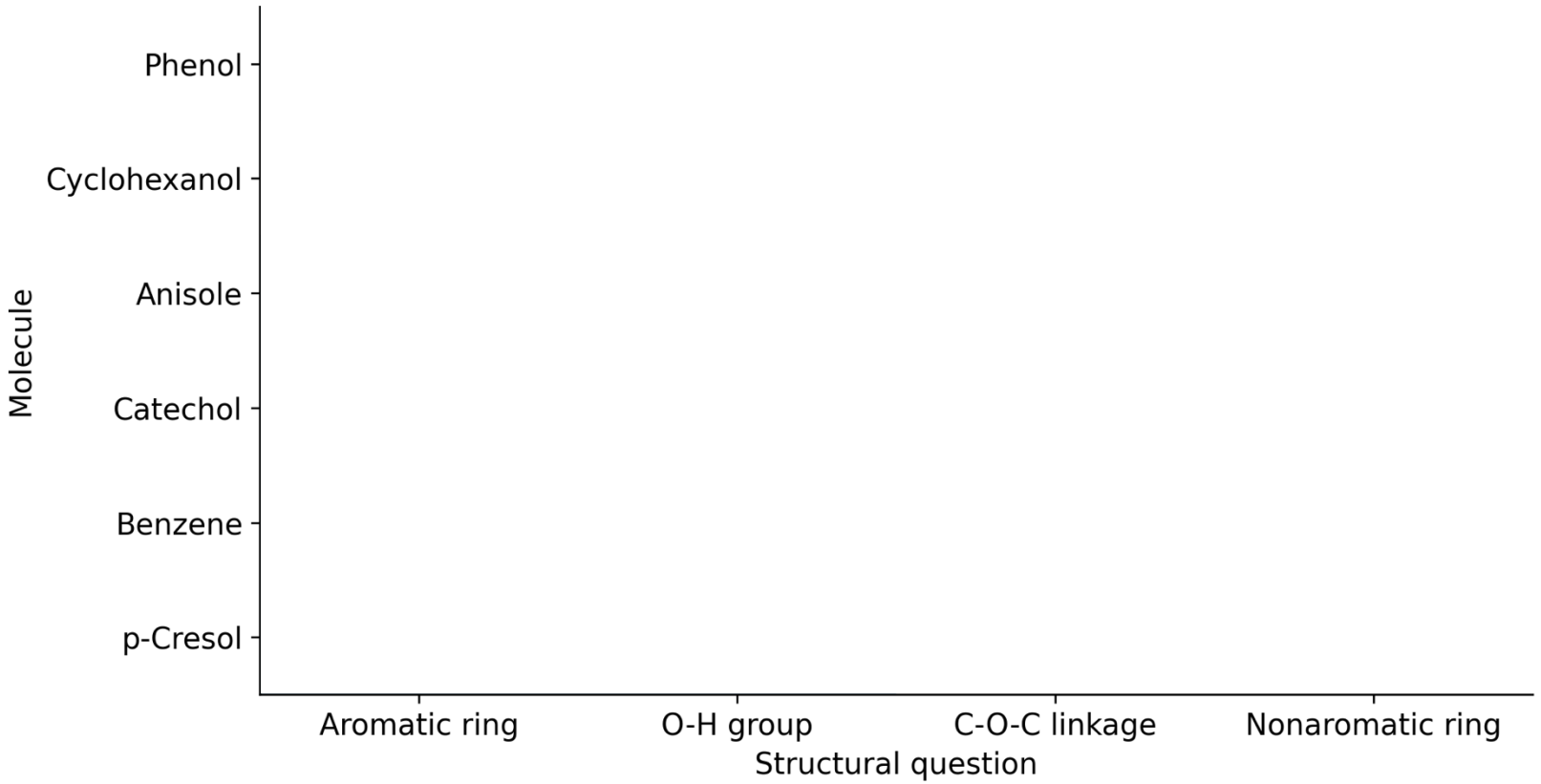


D: Benzene



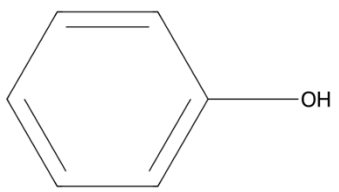
E: p-Cresol



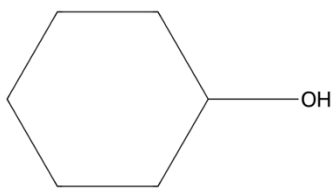


Different Fingerprints Capture Different Features

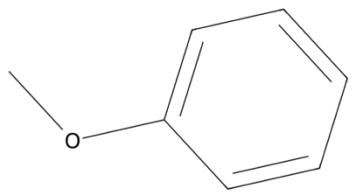
Reference: phenol



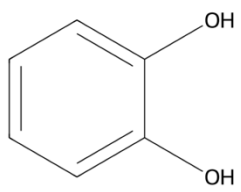
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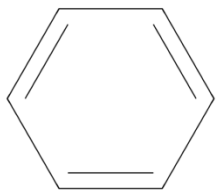
B: Anisole



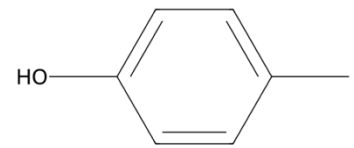
C: Catechol

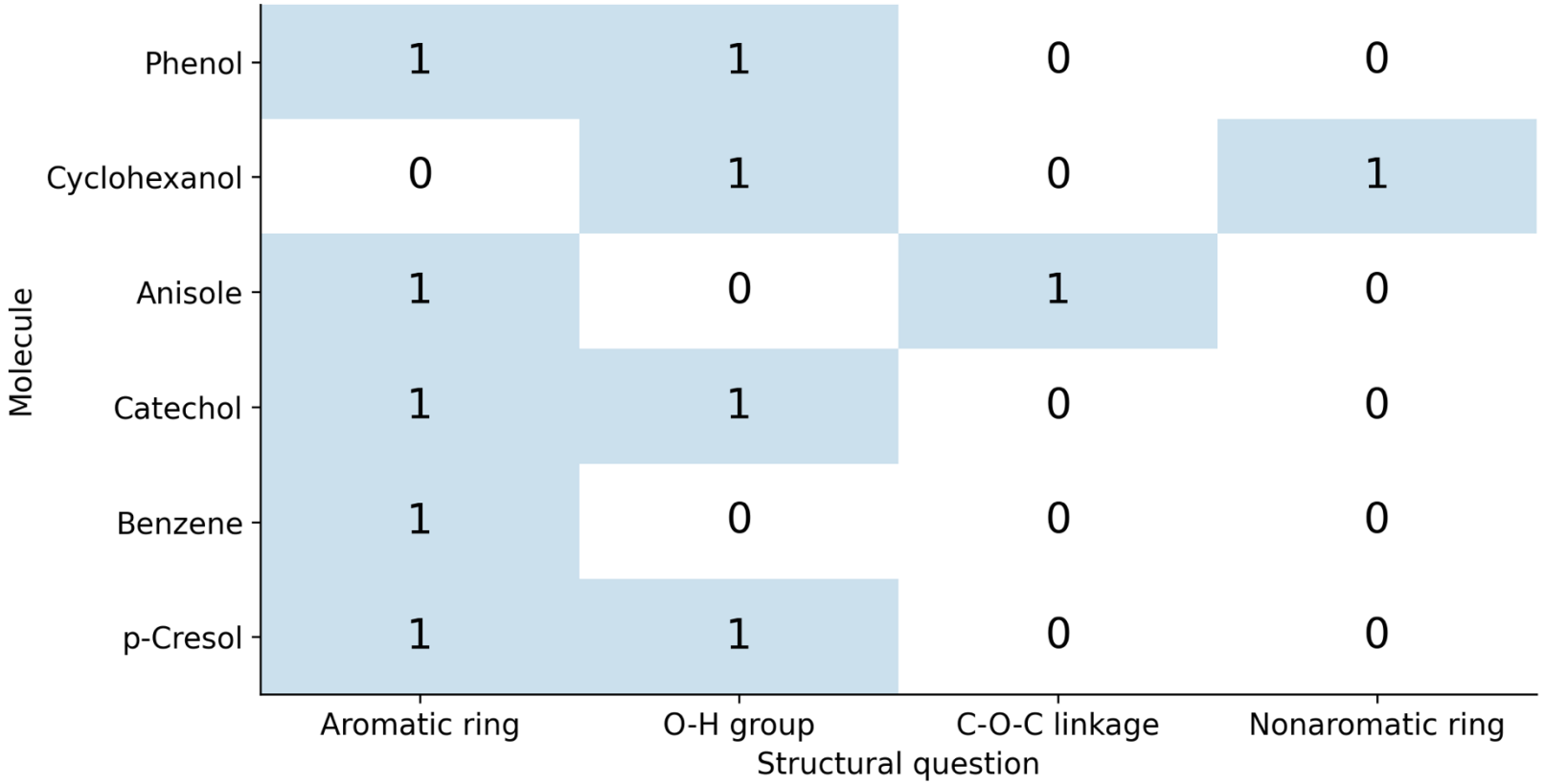


D: Benzene



E: p-Cresol





Molecule	O–H present?	Number of O–H groups
Phenol	1	1
Catechol	1	2

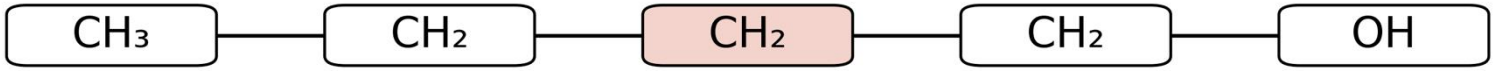
Binary features lose multiplicity. Count features recover it, but the feature definitions still determine what we can see.

Phenol, catechol, and p-cresol all receive [1,1,0,0]

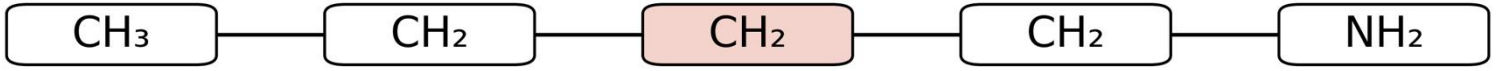
Morgan Fingerprints: Give Each Atom an Initial Label

- Begin with an atom description: element, attached hydrogens, charge, and other local attributes.

1-Butanol Red = selected center



Butan-1-amine Red = selected center

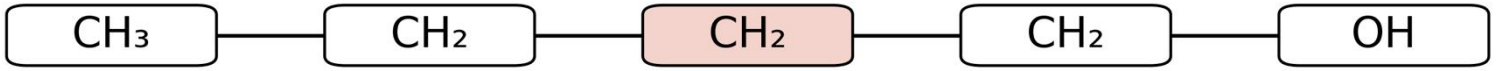


Selected center: initial attributes	1-Butanol	Butan-1-amine
Atomic number		
Neighbors, including H		
Attached H atoms		
Formal charge		
In a ring?		

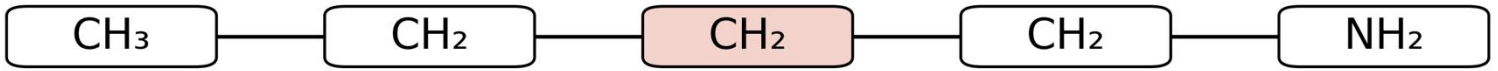
Morgan Fingerprints: Give Each Atom an Initial Label

- Begin with an atom description: element, attached hydrogens, charge, and other local attributes.
- The two highlighted CH_2 centers receive the same initial label.

1-Butanol Red = selected center



Butan-1-amine Red = selected center



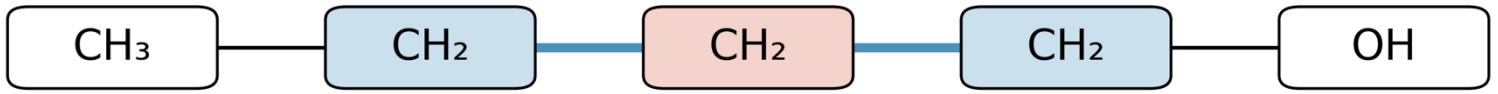
Selected center: initial attributes	1-Butanol	Butan-1-amine
Atomic number	6	6
Neighbors, including H	4	4
Attached H atoms	2	2
Formal charge	0	0
In a ring?	No	No

Same starting label: C, two attached H atoms, two heavy-atom neighbors, and zero charge.

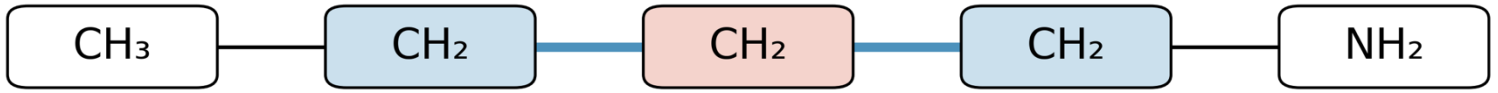
Radius 1: Look One Bond Away

- Update the center's label using its own label, its neighbors' labels, and the connecting bond types.
- Both centers connect to two CH_2 neighbors through single bonds.

1-Butanol Radius 1: center + atoms within 1 bond



Butan-1-amine Radius 1: center + atoms within 1 bond



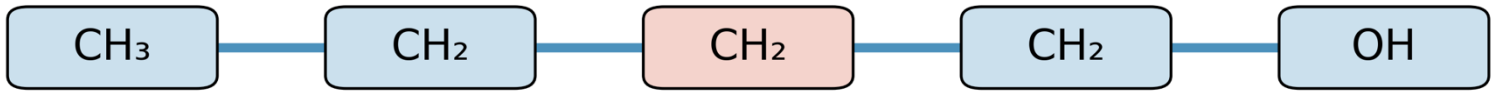
The selected centers still have the same radius-1 feature.

Radius 2: The Difference Reaches the Center

- Repeat the update: neighboring labels now contain information from farther away.
- The selected neighborhood reaches O in 1-butanol and N in butan-1-amine.

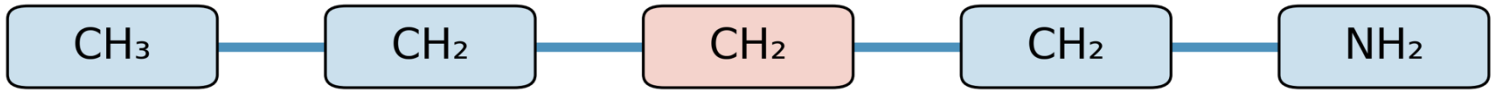
1-Butanol

Radius 2: center + atoms within 2 bonds



Butan-1-amine

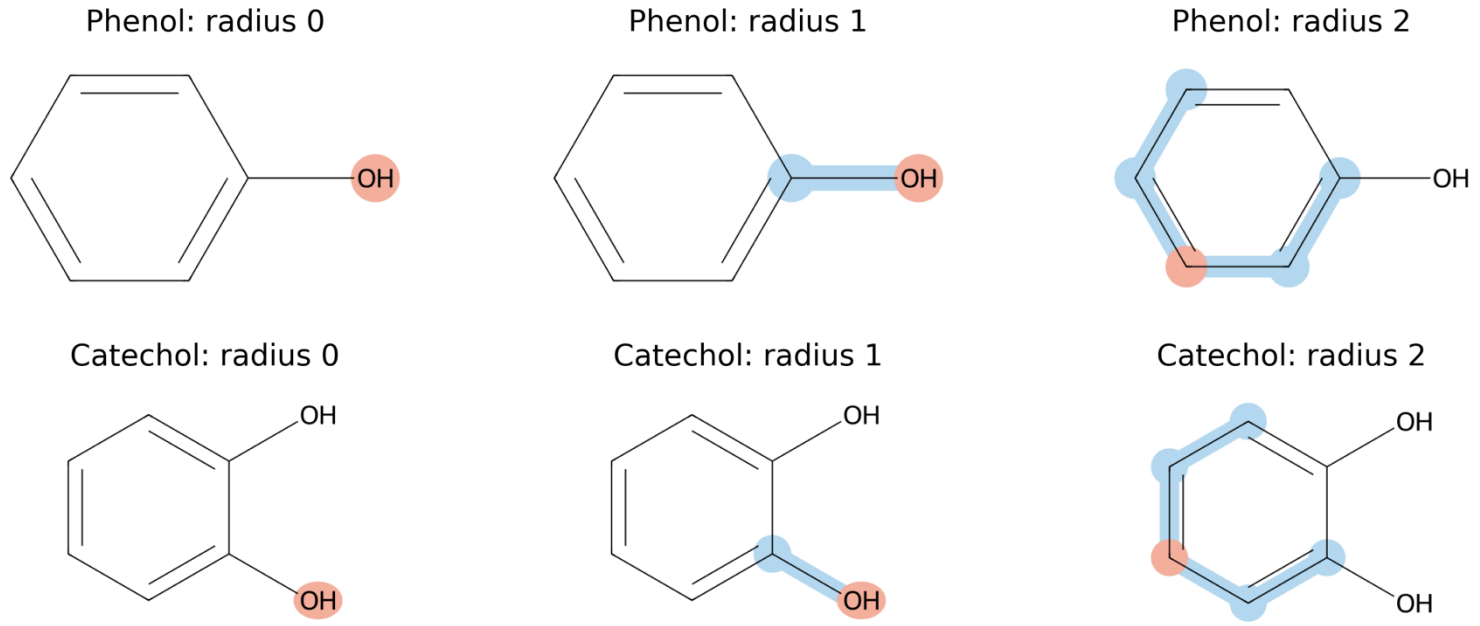
Radius 2: center + atoms within 2 bonds



The selected centers now have different radius-2 features.
The O vs. N difference reaches the red carbon.

Collect Neighborhood Features Across the Molecule

- Generate local features around atom centers, expanding up to the chosen radius.
- Pool the features into a molecular fingerprint.

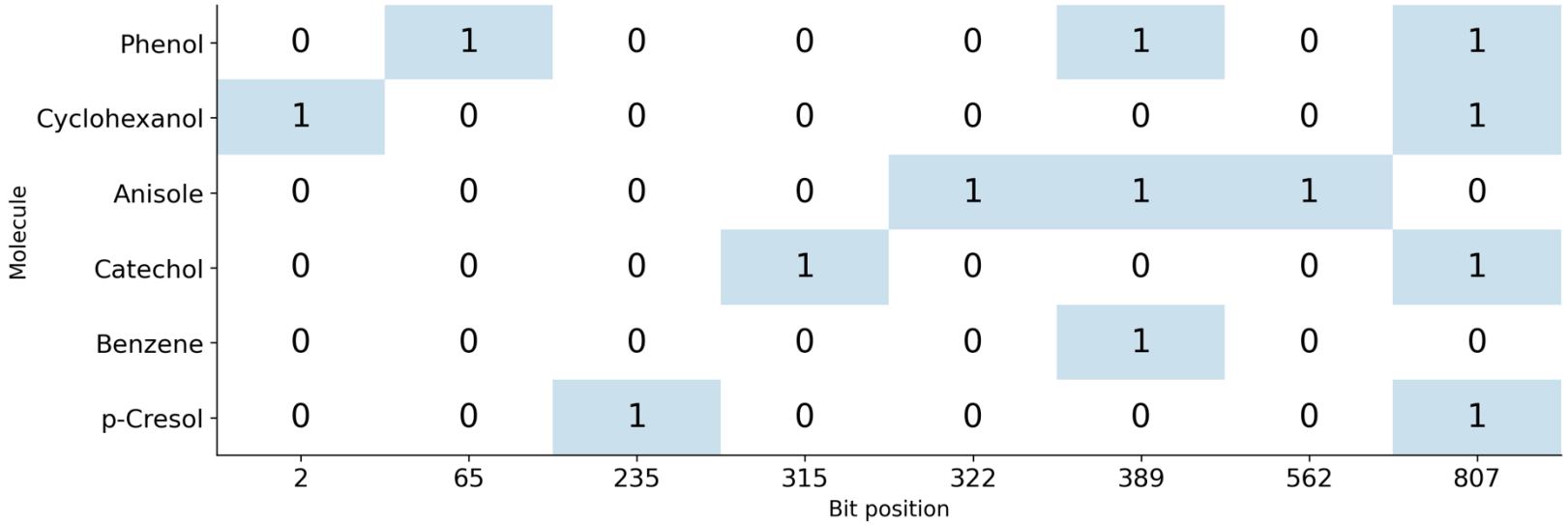


These are three actual shared Morgan features in phenol and catechol, shown at radii 0, 1, and 2. Each column shows an example of a feature these molecules share. The fingerprint collects many such local features across the molecule.

From Neighborhood Features to a Numerical Vector

Map each feature identifier into one of 2048 bit positions.
Set that bit to 1 when the feature occurs.

Phenol O atom with one H (radius 0) → Feature identifier 864662311 → Identifier % 2048 = 807 → Set bit 807 to 1

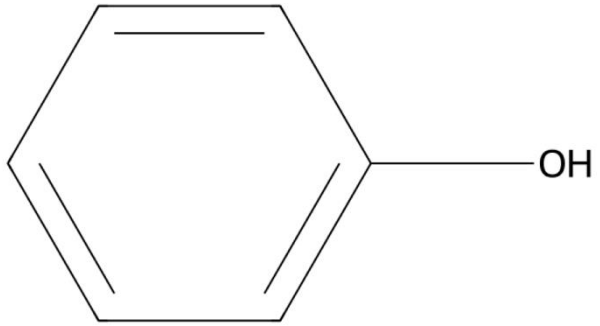


- *A molecule can activate many bits.*
- *Different features can map to the same bit: a hash collision.*

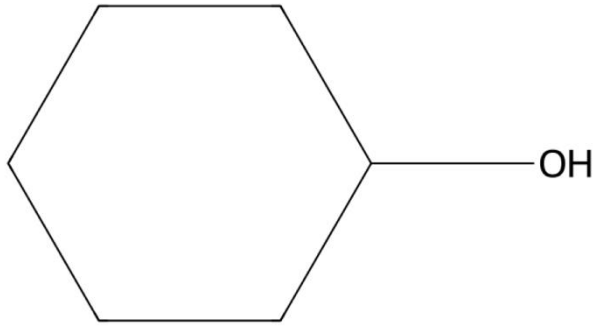
99% Matching Bits. Are These Molecules Similar?

- *Phenol and cyclohexanol each have 11 active bits; only one is shared.*

Phenol



Cyclohexanol



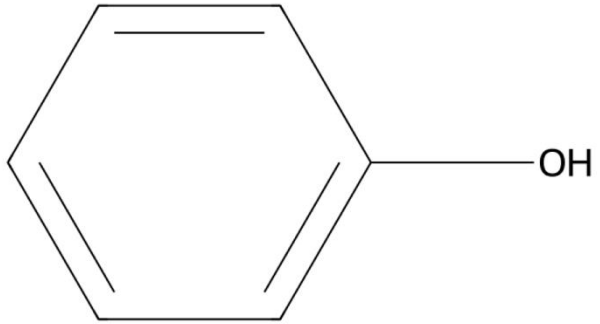
Comparison	Bit positions
Both 0	2027
Both 1	1
Different values	20

What is the bit-value agreement?

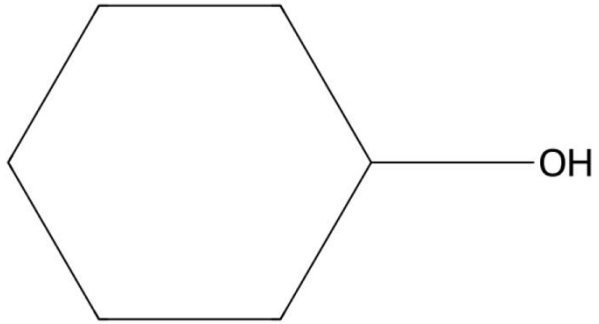
99% Matching Bits. Are These Molecules Similar?

- *Phenol and cyclohexanol each have 11 active bits; only one is shared.*
- *Their 2027 shared zeros inflate the percentage of matching bit values.*

Phenol



Cyclohexanol



Comparison	Bit positions
Both 0	2027
Both 1	1
Different values	20

What is the Tanimoto similarity?

Similarity for Fingerprints and Sets

To perform clustering, we must have a measure of similarity or distance between data points

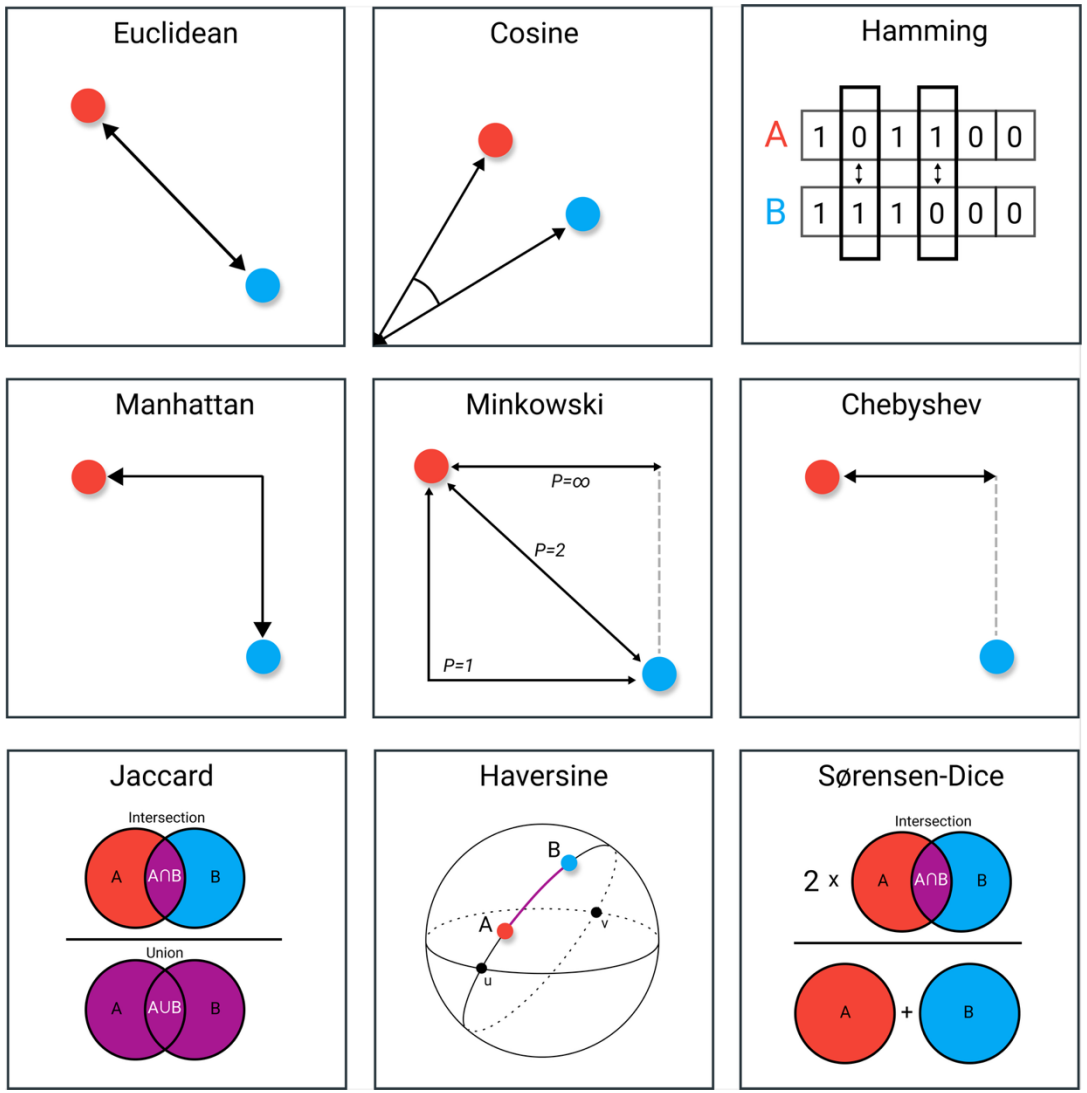
Tanimoto similarity

$$T(\mathbf{x}_i, \mathbf{x}_j) = \frac{\mathbf{x}_i^T \mathbf{x}_j}{\|\mathbf{x}_i\|^2 + \|\mathbf{x}_j\|^2 - \mathbf{x}_i^T \mathbf{x}_j}$$

For binary fingerprints, Tanimoto similarity is equivalent to Jaccard similarity.

$$\mathbf{x}_i = [1, 0, 1, 1, 0]$$
$$\mathbf{x}_j = [1, 1, 1, 0, 0]$$

$$T = \frac{2}{4} = 0.5$$

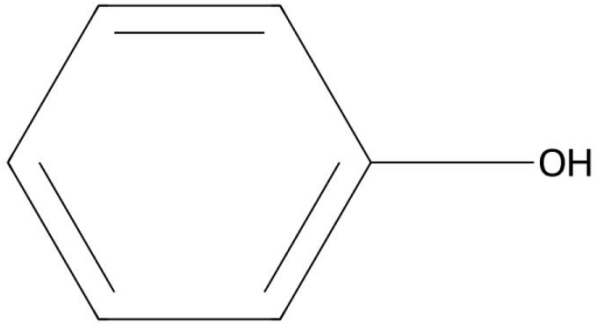


Maarten Grootendorst @ medium.com

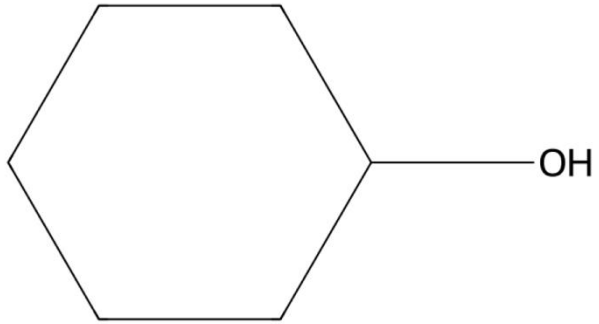
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- *Their 2027 shared zeros inflate the percentage of matching bit values.*

Phenol



Cyclohexanol



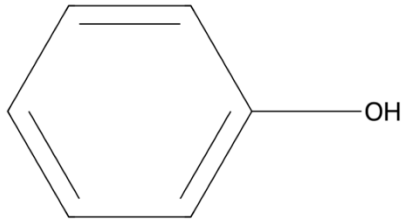
Comparison	Bit positions
Both 0	2027
Both 1	1
Different values	20

$$T = \frac{1}{11 + 11 - 1} = \frac{1}{21} = 0.048$$

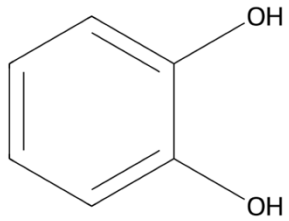
Which Candidate Is Closest to Phenol?

- *Compare all candidates using the same Morgan settings.*
  - *Rank candidates by decreasing Tanimoto similarity.*

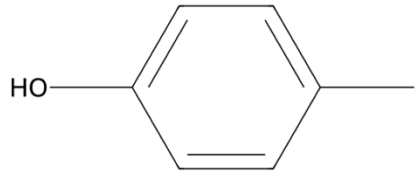
Reference: phenol



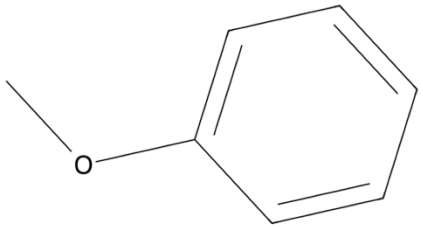
Catechol
Tanimoto = 0.500



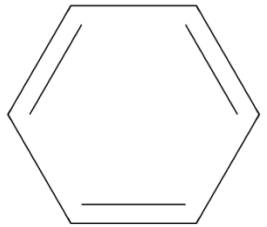
p-Cresol
Tanimoto = 0.412



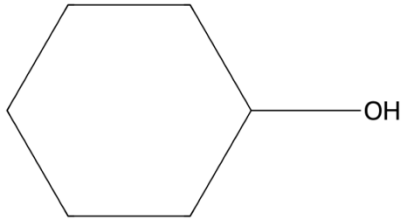
Anisole
Tanimoto = 0.316



Benzene
Tanimoto = 0.273



Cyclohexanol
Tanimoto = 0.048



Among these five candidates, catechol ranks first: T = 0.500.

Morgan radius = 2; fingerprint length = 2048 bits; chirality enabled.

Interactive Python Time

25_molecular_fingerprints.ipynb

In this lecture, we learned

- *Morgan fingerprints and local atomic environments*
- *Generating fingerprints with RDKit*
- *Comparing molecules using fingerprint similarity*

Next lecture:

- *More molecular fingerprints*
- *Molecular descriptors*